



A PRACTICAL, SYSTEMATIC APPROACH

ECG READING GUIDE

UNDERSTANDING AND INTERPRETING ECGs

FOR

Medical Students • Nursing Students • Pharmacists • Paramedical &
Emergency-Care Learners

Medical Reference Series | 2026

Disclaimer

This guide is intended for education and revision only. It is designed to help healthcare students and professionals understand the basic principles of electrocardiography and develop a systematic approach to ECG interpretation.

An ECG must always be interpreted together with the patient's symptoms, clinical examination, medical history, medications, laboratory findings, and other investigations where appropriate.

The information in this guide should not be used as a substitute for diagnosis, treatment decisions, emergency medical care, or advice from a qualified medical practitioner or cardiologist.

⚠ WARNING

- ECG findings can be nonspecific, evolving, technically distorted, or normal despite clinically important disease.
- When an acute cardiac condition is suspected, appropriate clinical evaluation should not be delayed because of a single ECG result.

How This Guide Is Organized

This guide builds ECG interpretation in four stages — from the raw electrical signal, to the normal tracing, to recognising abnormal patterns, to a complete, repeatable interpretation framework.

Part	Focus
I — Building the Foundation	What an ECG is, how the signal is generated, the paper grid, and the 12 leads
II — Learn the Normal ECG	Waves, segments, intervals, rate calculation, and a systematic normal check
III — Recognising Abnormal Patterns	Rhythm disturbances, AV/bundle branch blocks, ischemia/infarction, electrolytes
IV — Interpret Like a System	The full reading algorithm, common traps, exam revision, and a rapid checklist

The Core Sequence

Every systematic ECG read in this guide follows the same order, repeated in Chapters 7 and 12:

Technical Check

- ↓
- Rate
- ↓
- Rhythm
- ↓
- P
- ↓
- PR
- ↓
- QRS
- ↓
- Axis
- ↓

- ST
- ↓
- T
- ↓
- QT/QTc
- ↓
- Lead Comparison
- ↓
- Final Impression

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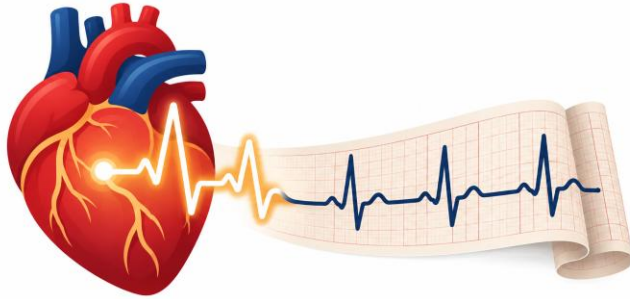
PART I

Building the Foundation

*What an ECG records, how the electrical signal is generated,
the paper grid, and the 12-lead system*

CHAPTER 1

Understanding the ECG



Before trying to recognise myocardial infarction, atrial fibrillation or heart block, understand one simple idea:

“An ECG is a surface recording of the heart's electrical activity over time.”

The heart is a mechanical pump, but its contractions are coordinated by electrical activity. As electrical impulses travel through the atria and ventricles, differences in electrical potential can be detected by electrodes placed on the body. The ECG machine converts these signals into waveforms.

1.1 What Does ECG Stand For?

- E — Electro: electrical activity
- C — Cardio: heart
- G — Gram: recorded representation

So an electrocardiogram is a recording of cardiac electrical activity. You may also encounter the abbreviation EKG, derived from the

German spelling Elektrokardiogramm. ECG and EKG refer to the same test.

1.2 What Information Can an ECG Provide?

A correctly recorded and interpreted ECG can provide clues about:

- Heart rate
- Cardiac rhythm
- Impulse conduction
- AV conduction abnormalities
- Bundle branch conduction
- Myocardial ischemia or infarction patterns
- Chamber enlargement/hypertrophy patterns
- Certain electrolyte disturbances
- Effects of selected medications
- Pacemaker activity
- Some causes of syncope or palpitations

★ KEY POINT

- An ECG finding is usually one piece of the clinical picture, not the entire diagnosis.

1.3 What an ECG Does NOT Directly Show

This distinction is important for beginners. An ECG primarily evaluates electrical activity. It does not directly provide detailed images of:

- Heart valves
- Cardiac anatomy
- Ventricular wall motion
- Ejection fraction

- Coronary artery lumen
- Structural defects

These questions may require investigations such as echocardiography, coronary imaging, CT, MRI, or other appropriate tests.

 TIP

- ECG → electrical information. Echocardiography → structural and functional imaging.

1.4 Why Is an ECG Ordered?

Common reasons include evaluation of:

Presentation	Why an ECG Is Ordered
Chest discomfort	To look for patterns compatible with myocardial ischemia/infarction and other cardiac abnormalities
Palpitations	To identify rhythm disturbances
Syncope or dizziness	To look for bradyarrhythmias, tachyarrhythmias, conduction abnormalities and other possible cardiac causes
Shortness of breath	May provide evidence supporting a cardiac cause or associated abnormalities
Abnormally fast or slow pulse	To determine the underlying rhythm
Suspected electrolyte disturbance	Some electrolyte abnormalities produce characteristic ECG patterns
Medication monitoring	Certain medications can affect conduction, rhythm, QRS morphology, ST-T appearance or QT interval
Preoperative assessment	Obtained when clinically indicated according to age, risk factors, procedure and local protocols

Chapter Summary

An ECG is a surface recording of cardiac electrical activity, not a structural image. It can reveal rate, rhythm, conduction, ischemic

patterns, and some electrolyte or medication effects, but questions about anatomy, wall motion, or ejection fraction need imaging such as echocardiography instead.

Key Takeaways

- ECG and EKG are the same test, named from the English and German spellings respectively.
- An ECG finding is one piece of the clinical picture, not a standalone diagnosis.
- An ECG shows electrical activity, not anatomy, wall motion, or ejection fraction.
- Chest pain, palpitations, syncope, and suspected electrolyte disturbance are among the most common reasons an ECG is ordered.

✓ QUICK RECALL

- What do the three letters E, C, and G stand for?
- Name three things an ECG cannot directly show.
- Why might an ECG be ordered for a patient with syncope?

CHAPTER 2

From Electrical Impulse to ECG Waveform

ECG interpretation becomes much easier when you understand where the electrical signal travels. Think of the conduction pathway as a sequence:

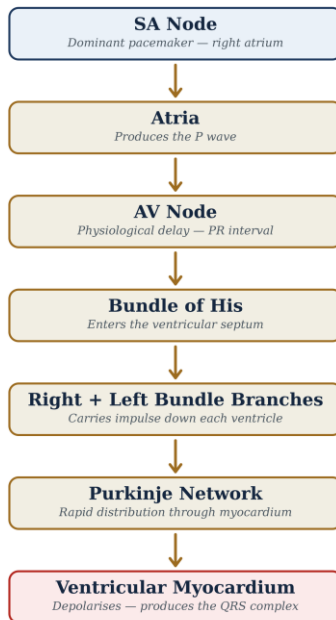


Figure 2.1 — The cardiac conduction pathway, from the SA node to the ventricular myocardium.

This electrical sequence produces the major components visible on an ECG.

2.1 SA Node — The Usual Starting Point

The sinoatrial (SA) node is located in the right atrium and normally acts as the heart's dominant pacemaker. Its spontaneous electrical impulses initiate normal sinus beats.

★ KEY POINT

- A normal sinus rhythm in an adult generally has a rate of approximately 60–100 beats/min — but rate alone does not establish sinus rhythm. You must also assess the P waves and their relationship to the QRS complexes.

2.2 Atrial Depolarisation

Once the impulse leaves the SA node, electrical activation spreads through the atrial myocardium. This electrical event produces the P wave. Therefore:

“P wave = atrial depolarisation”

Mechanical atrial contraction follows the electrical activation.

2.3 The AV Node — A Useful Delay

The impulse then reaches the atrioventricular (AV) node. Conduction slows here. This physiological delay allows time for ventricular filling before ventricular activation occurs. AV conduction contributes to what we assess using the PR interval — measured from the beginning of the P wave to the beginning of the QRS complex.

★ KEY POINT

- Typical adult PR interval reference: 0.12–0.20 seconds.

2.4 His-Purkinje System

After passing through the AV node, the impulse travels through the Bundle of His, then the bundle branches, then the Purkinje fibres. This specialised conduction system distributes electrical activation rapidly throughout the ventricles. The resulting ventricular depolarisation produces the QRS complex.

QRS Duration	Interpretation
< 0.12 sec	Considered narrow
≥ 0.12 sec	Considered wide — requires further interpretation

2.5 Ventricular Recovery

After depolarisation, ventricular cells return toward their resting electrical state. This process is called ventricular repolarisation. Its most recognisable ECG representation is the T wave.

✓ QUICK RECALL

- P → atria depolarise. QRS → ventricles depolarise. T → ventricles repolarise.

Chapter Summary

The cardiac conduction system runs a fixed route — SA node, atria, AV node, Bundle of His, bundle branches, Purkinje network, ventricular myocardium — and each stop on that route produces a specific, recognisable part of the ECG waveform: the P wave, the PR interval, and the QRS complex, followed by the T wave as the ventricles recover.

Key Takeaways

- The SA node is the heart's normal dominant pacemaker, firing at approximately 60–100 bpm.
- The AV node's conduction delay is what the PR interval (0.12–0.20 sec) measures.
- QRS duration below 0.12 sec is narrow; 0.12 sec or above is wide and needs further interpretation.
- The T wave represents ventricular repolarisation, not a second depolarisation event.

CHAPTER 3

Reading the ECG Grid

Many interpretation errors start before the waveform is even analysed.

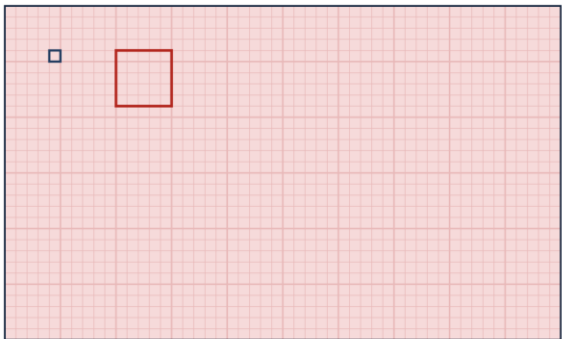
⚠ WARNING

- Always check paper speed and calibration before measuring rate, intervals, or voltage.

3.1 Standard Paper Speed

A commonly used standard ECG paper speed is 25 mm/second. At this speed:

25 mm/second — Standard ECG Paper Speed



VOLTAGE (v)

- 10 mm = 1 mV
- 1 small square (1 mm) = 0.1 mV
- 1 large square (5 mm) = 0.5 mV

— 1 small square = 0.04 sec

— 1 large square (5 small) = 0.20 sec

Figure 3.1 — Standard ECG grid: time runs horizontally, voltage runs vertically.

Measurement	Value
1 small square	0.04 sec

1 large square	0.20 sec
5 large squares	1 second
30 large squares	6 seconds

These numbers form the basis of most basic ECG calculations.

3.2 Voltage Calibration

With standard calibration, 10 mm = 1 mV. Therefore vertically:

Measurement	Value
1 small square (1 mm)	0.1 mV
1 large square (5 mm)	0.5 mV

⚠ WARNING

- Never assume the recording uses standard settings without checking the tracing.

3.3 Horizontal vs Vertical

Think of the ECG grid as having two dimensions.

Horizontal = Time

Used for: heart rate, PR interval, QRS duration, QT interval.

Vertical = Voltage

Used for: wave amplitude, ST displacement, voltage criteria.

✓ QUICK RECALL

- Memory rule — Across = Time. Up/Down = Voltage.

Chapter Summary

Every ECG measurement depends on the paper speed and calibration being what you assume they are. At the standard 25 mm/sec and 10 mm/mV settings, the horizontal axis reads time and the vertical axis reads voltage — and confirming both before measuring anything is the first step of every systematic read.

Key Takeaways

- Standard paper speed is 25 mm/sec, where one small square = 0.04 sec and one large square = 0.20 sec.
- Standard calibration is 10 mm = 1 mV, where one large square vertically = 0.5 mV.
- Horizontal measurements are time-based; vertical measurements are voltage-based.
- Always confirm the tracing's actual speed and calibration before measuring — never assume standard settings.

CHAPTER 4

Understanding the 12-Lead ECG

One of the most common beginner mistakes is thinking that a 12-lead ECG uses 12 electrodes.

★ KEY POINT

- A standard 12-lead ECG normally uses 10 electrodes to generate 12 leads — 4 limb electrodes and 6 chest electrodes.

A lead is an electrical viewpoint of the heart. An electrode is the physical sensor attached to the patient.

4.1 The 12 Leads

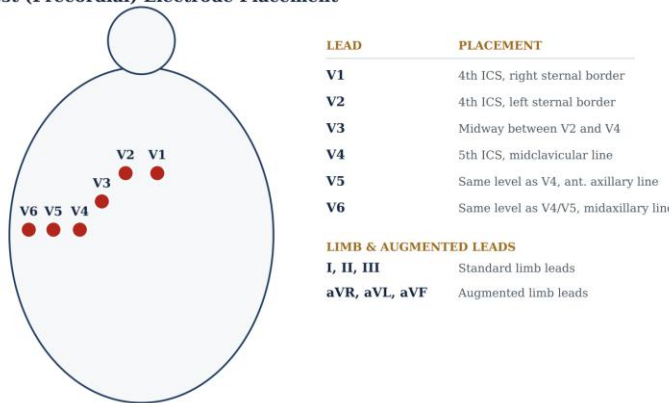
Group	Leads
Standard limb leads	I, II, III
Augmented limb leads	aVR, aVL, aVF
Precordial / chest leads	V1, V2, V3, V4, V5, V6

Together they provide different electrical perspectives.

4.2 Chest Electrode Placement

Correct placement matters because incorrect positioning can create misleading patterns.

Chest (Precordial) Electrode Placement



Practical rule: locate V4 accurately before placing V3.

Figure 4.1 — Standard chest (precordial) electrode placement, V1–V6.

Lead	Placement
V1	Fourth intercostal space, right sternal border
V2	Fourth intercostal space, left sternal border
V3	Midway between V2 and V4
V4	Fifth intercostal space, midclavicular line

V5	Same horizontal level as V4, anterior axillary line
V6	Same horizontal level as V4/V5, midaxillary line

 **TIP**

- Practical rule: locate V4 accurately before placing V3.

Chapter Summary

A standard 12-lead ECG is built from only 10 physical electrodes — 4 on the limbs and 6 on the chest — combined to generate 12 distinct electrical viewpoints. Correct chest electrode placement, especially locating V4 before V3, is essential because misplacement can create misleading waveform patterns.

Key Takeaways

- 12 leads come from only 10 electrodes: 4 limb + 6 chest.
- The 12 leads split into standard limb (I, II, III), augmented limb (aVR, aVL, aVF), and precordial (V1–V6) groups.
- V1 and V2 sit at the 4th intercostal space on either side of the sternum; V4 sits at the 5th intercostal space, midclavicular line.
- Locate V4 first, then place V3 midway between V2 and V4.

✓ **QUICK RECALL**

- How many electrodes produce a standard 12-lead ECG?
- What is the placement of V1 and V4?
- Why should V4 be located before V3?

PART II

Learn the Normal ECG

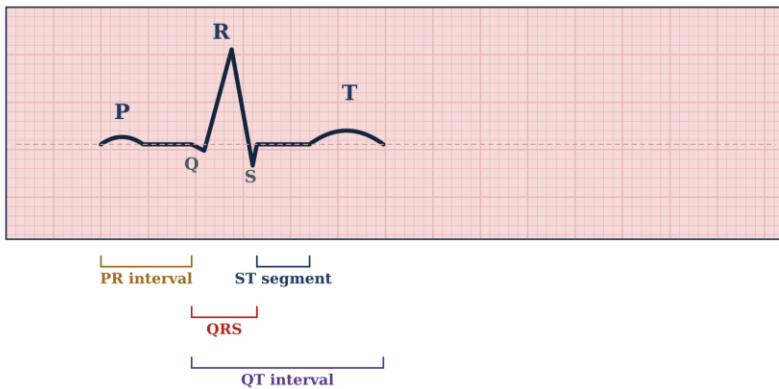
*Waves, segments and intervals, calculating heart rate,
and a systematic check of the normal tracing*

CHAPTER 5

Waves, Segments and Intervals

A complete cardiac cycle contains several waves, intervals and segments. Instead of memorising their shapes alone, connect each one with the underlying electrical event.

Anatomy of One Cardiac Cycle



P = atrial depolarisation • *QRS* = ventricular depolarisation • *T* = ventricular repolarisation

Figure 5.1 — Anatomy of one cardiac cycle: P wave, QRS complex, T wave, and the PR, QRS, ST and QT measurements.

5.1 P Wave

Represents atrial depolarisation. Typical features include:

- Smooth morphology
- Usually upright in lead II during sinus rhythm
- Duration generally < 0.12 sec

- Amplitude generally ≤ 2.5 mm in limb leads

P-wave morphology can provide clues about atrial activation and enlargement.

5.2 PR Interval

Measured from the start of the P wave to the start of the QRS. Typical reference range: 0.12–0.20 sec.

A prolonged PR interval can occur with delayed AV conduction. A short PR interval requires interpretation in the context of the complete ECG and clinical setting.

5.3 QRS Complex

Represents ventricular depolarisation.

Term	Meaning
Q wave	First negative deflection before an R wave
R wave	First positive deflection
S wave	Negative deflection following an R wave

A QRS ≥ 0.12 sec is considered wide and may be seen with conditions such as bundle branch block, ventricular rhythms, pacing, or other conduction abnormalities.

5.4 ST Segment

The ST segment follows ventricular depolarisation and is assessed relative to the ECG baseline.

ST Elevation

Possible causes include acute myocardial injury patterns, pericarditis, early repolarisation and other conditions.

ST Depression

Possible causes include ischemia, reciprocal changes, rate-related changes, medication effects and other abnormalities.

⚠ WARNING

- ST elevation by itself does not automatically equal myocardial infarction. Lead distribution, morphology, clinical presentation, serial ECGs and appropriate biomarkers matter.

5.5 T Wave

Primarily represents ventricular repolarisation. Changes in T-wave morphology can occur with:

- Ischemia
- Electrolyte abnormalities
- Ventricular strain
- Conduction abnormalities
- CNS conditions
- Normal variants

⚠ WARNING

- Never diagnose solely from an isolated T-wave abnormality.

5.6 QT Interval and QTc

The QT interval extends from the beginning of the QRS to the end of the T wave. Because QT duration varies with heart rate, clinicians commonly use a corrected value, QTc.

★ KEY POINT

- A markedly prolonged QT/QTc can increase susceptibility to potentially dangerous ventricular arrhythmias. Medication history and electrolyte abnormalities should therefore be considered when evaluating QT prolongation.

Chapter Summary

Each part of one cardiac cycle maps to a specific electrical event and has its own typical duration or amplitude range — the P wave for atrial depolarisation, the PR interval for AV delay, the QRS for ventricular depolarisation, the ST segment and T wave for repolarisation, and the QT/QTc as an overall marker of repolarisation duration that is sensitive to medications and electrolytes.

Key Takeaways

- Normal P-wave duration is under 0.12 sec, with amplitude up to about 2.5 mm in the limb leads.
- PR interval reference range is 0.12–0.20 sec; QRS is narrow below 0.12 sec and wide at or above it.
- ST elevation and depression each have multiple possible causes — neither is diagnostic on its own.
- QT/QTc prolongation matters because it can predispose to dangerous ventricular arrhythmias.

✓ QUICK RECALL

- What electrical event does the P wave represent? The QRS? The T wave?
- What is the normal PR interval range, and what QRS duration is considered wide?
- Why is ST elevation alone not sufficient to diagnose myocardial infarction?

CHAPTER 6

Rate and Rhythm Made Simple

Before calculating rate, answer one question: is the rhythm regular or irregular? The method you choose depends on this.

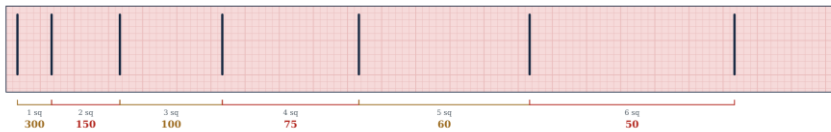
6.1 Method 1 — The 300 Method

Useful for a reasonably regular rhythm at 25 mm/sec. Count the large squares between consecutive R waves.

★ KEY POINT

- Formula: Heart Rate $\approx 300 \div$ number of large squares between R waves.

The “300 Method” — Counting Large Squares Between R Waves



Formula: Heart Rate $\approx 300 \div$ number of large squares between consecutive R waves

Figure 6.1 — The 300 method: large squares between R waves and the corresponding heart rate.

A useful sequence to remember is: 300 – 150 – 100 – 75 – 60 – 50.

6.2 Method 2 — The 1500 Method

For a regular rhythm: Heart Rate $\approx 1500 \div$ small squares between R waves.

🔪 TIP

- Example: 20 small squares between two R waves $\rightarrow 1500 \div 20 = 75$ bpm.

6.3 Method 3 — Six-Second Method

Useful for estimating the ventricular rate in irregular rhythms. At 25 mm/sec, 30 large squares = 6 seconds. Count the QRS complexes in that interval and multiply by 10.

TIP

- Example: 9 QRS complexes in six seconds $\rightarrow 9 \times 10 =$ approximately 90 bpm. This is an estimate rather than a precision measurement.

Chapter Summary

Heart rate can be calculated three ways depending on rhythm regularity: the 300 method and 1500 method both work for a regular rhythm by counting squares between R waves, while the six-second method — counting QRS complexes across 30 large squares and multiplying by 10 — is the practical choice for an irregular rhythm.

Key Takeaways

- Always determine whether the rhythm is regular or irregular before choosing a rate-calculation method.
- The 300 sequence — 300, 150, 100, 75, 60, 50 — is worth memorising for quick regular-rhythm estimates.
- The six-second method is the practical choice for irregular rhythms, but it is an estimate, not a precise measurement.

QUICK RECALL

- What is the formula for the 300 method?
- How many large squares correspond to six seconds at standard paper speed?
- Which rate-calculation method is best suited to an irregular rhythm?

CHAPTER 7

A Systematic Normal ECG Check



⚠ WARNING

- Do not decide an ECG is normal just because the heart rate is normal. Use a sequence.

7.1 The Sequence

RATE



RHYTHM



P



PR



QRS



AXIS



ST



T



QT/QTc

7.2 What to Ask at Each Step

Step	Question
Rate	Appropriate?
Rhythm	Regular or irregular?
P waves	Present? Consistent? Relationship to QRS?
PR	Appropriate and constant?
QRS	Narrow or wide? Morphology?
Axis	Normal or deviated?
ST	Elevation or depression?
T	Expected polarity and morphology?
QT/QTc	Appropriate?

★ **KEY POINT**

- Only after checking the entire tracing should you form an interpretation.

Chapter Summary

A tracing is not confirmed normal just because the rate looks fine. The nine-step sequence — rate, rhythm, P, PR, QRS, axis, ST, T, QT/QTc — must be checked in full before any interpretation is offered, and this same sequence becomes the backbone of the complete reading algorithm in Chapter 12.

Key Takeaways

- A normal-looking rate does not by itself confirm a normal ECG.
- The nine-step sequence covers rate, rhythm, P, PR, QRS, axis, ST, T, and QT/QTc, in that order.
- Form an interpretation only after completing every step of the sequence.

PART III

Recognising Abnormal Patterns

*Rhythm disturbances, AV and bundle branch blocks,
ischemia / infarction, and electrolyte clues*

CHAPTER 8

Common Rhythm Abnormalities

Learning to recognise a handful of distinctive rhythm patterns covers most of what you will encounter in practice.

Recognising Rhythm Patterns at a Glance

Normal Sinus Rhythm — regular, narrow QRS, P before every QRS



Atrial Fibrillation — irregularly irregular, no consistent P waves



Ventricular Tachycardia — regular, wide QRS, rapid rate



Figure 8.1 — Normal sinus rhythm compared with atrial fibrillation and ventricular tachycardia.

8.1 Sinus Tachycardia

Typical pattern: sinus P waves, P before each QRS, consistent PR relationship, adult rate > 100 bpm.

Possible clinical associations include exercise, fever, pain, anxiety, hypovolemia, anemia, infection and other physiological or pathological causes.

★ **KEY POINT**

- The ECG identifies the rhythm; it does not tell you the cause by itself.

8.2 Sinus Bradycardia

Typical pattern: sinus P waves, P before each QRS, regular rhythm, adult rate < 60 bpm.

It may be physiological in some healthy individuals and athletes, or associated with medications, conduction disease, metabolic abnormalities and other conditions.

8.3 Atrial Fibrillation

Classic ECG features:

- No consistent discrete sinus P waves
- Irregularly irregular ventricular rhythm
- Variable R-R intervals

⚠ **WARNING**

- AF is clinically important partly because it may increase thromboembolic risk, but management depends on the patient's broader clinical circumstances.

8.4 Atrial Flutter

Atrial flutter may produce organised rapid atrial activity, classically described as flutter or “saw-tooth” waves. The ventricular response can be regular or variable depending on AV conduction.

8.5 Ventricular Tachycardia

A regular broad-complex tachycardia should raise serious concern for VT, particularly in the appropriate clinical context. Features may include:

- Rapid ventricular rate
- Wide QRS complexes
- AV dissociation in some cases
- Capture/fusion beats in some cases

⚠ WARNING

- A broad-complex tachycardia should be approached cautiously; morphology and clinical context are required for definitive interpretation.

8.6 Ventricular Fibrillation

VF produces chaotic ventricular electrical activity without an organised QRS rhythm.

⚠ WARNING

- VF is a cardiac-arrest rhythm requiring immediate resuscitation according to appropriate protocols.

Chapter Summary

Rhythm recognition rests on a small number of distinguishing features: rate and P-QRS relationship separate sinus tachycardia and bradycardia from each other; the absence of discrete P waves with an irregularly irregular rhythm marks atrial fibrillation; a regular wide-complex tachycardia raises concern for VT; and VF is a chaotic, pulseless, cardiac-arrest rhythm.

Key Takeaways

- Sinus tachycardia and bradycardia both keep a P before every QRS — only the rate and clinical context differ.
- Atrial fibrillation is defined by an irregularly irregular rhythm with no consistent P waves.
- A regular, wide-complex tachycardia should be treated as VT until proven otherwise.
- VF has no organised QRS rhythm and is a cardiac-arrest rhythm.

✓ QUICK RECALL

- What are the three classic ECG features of atrial fibrillation?
- What rate threshold defines sinus tachycardia in an adult?
- Why should a broad-complex tachycardia be approached cautiously?

CHAPTER 9

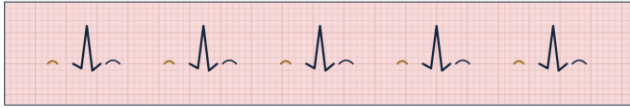
AV Blocks and Bundle Branch Blocks

The easiest way to understand AV block is to ask: “what is happening between P and QRS?”

AV Block Patterns — What Is Happening Between P and QRS?

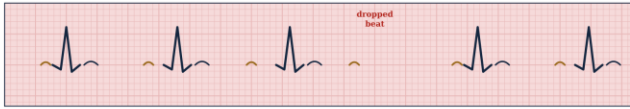
First-Degree AV Block

PR interval is prolonged (>0.20 sec) but constant — every P wave is followed by a QRS.



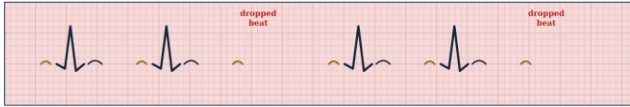
Second-Degree AV Block — Mobitz I (Wenckebach)

PR interval lengthens progressively until a P wave is not followed by a QRS, then the cycle repeats.



Second-Degree AV Block — Mobitz II

PR interval stays constant in conducted beats, but some P waves are suddenly not followed by a QRS.



Third-Degree AV Block (Complete Heart Block)

Atrial and ventricular activity are fully dissociated — P waves and QRS complexes each follow their own independent rhythm.



Figure 9.1 — First-degree, Mobitz I, Mobitz II, and third-degree (complete) AV block.

9.1 First-Degree AV Block

Typical ECG feature: PR interval > 0.20 sec, but every atrial impulse is conducted to the ventricles.

★ KEY POINT

- Think: PR is long, but no QRS is dropped.

9.2 Second-Degree AV Block — Mobitz I

Also called Wenckebach. Classic pattern: PR progressively lengthens until a P wave is not followed by a QRS complex. Then the sequence begins again.

★ KEY POINT

- Memory concept: Longer → Longer → Drop.

9.3 Second-Degree AV Block — Mobitz II

PR intervals in conducted beats are typically constant, but some P waves are unexpectedly not followed by QRS complexes.

⚠ WARNING

- This pattern is clinically important because it may reflect disease within the His-Purkinje conduction system and can progress to higher-grade block.

9.4 Third-Degree AV Block

Also called complete heart block. Atrial and ventricular electrical activity proceed independently. Therefore: P waves have their own rhythm, while QRS complexes have another rhythm. This is AV dissociation.

The ventricular escape rhythm may be slow and can be narrow or wide depending on its origin.

Chapter Summary

AV block severity is read from the relationship between P waves and QRS complexes: a constant but long PR in first-degree block, a progressively lengthening PR ending in a dropped beat in Mobitz I, a sudden dropped beat with otherwise constant PR in Mobitz II, and complete independence between P waves and QRS complexes in third-degree block.

Key Takeaways

- First-degree block: PR > 0.20 sec, but every P is still conducted.
- Mobitz I (Wenckebach): PR lengthens progressively, then one P wave is dropped, then the cycle restarts.
- Mobitz II: PR is constant in conducted beats, but some P waves are suddenly dropped — clinically more concerning than Mobitz I.
- Third-degree (complete) block: P waves and QRS complexes are fully dissociated, each with its own independent rhythm.

✓ QUICK RECALL

- What distinguishes Mobitz I from Mobitz II?
- What PR interval value defines first-degree AV block?
- What does AV dissociation mean in third-degree block?

CHAPTER 10

Ischemia and Myocardial Infarction

This is one area where oversimplified ECG rules can be dangerous.

⚠ WARNING

- Do not use “ST elevation = heart attack” as a standalone diagnostic rule.

Instead ask: WHERE are the changes? HOW MANY contiguous leads? WHAT is the morphology? ARE reciprocal changes present? WHAT symptoms does the patient have? HOW has the ECG changed over time?

10.1 Important ECG Changes

Acute coronary syndromes may be associated with:

- ST elevation
- ST depression
- T-wave inversion
- Hyperacute T waves
- Pathological Q waves
- Reciprocal changes

But not every patient follows a textbook sequence.

10.2 Think in Anatomical Territories

12-Lead Layout and Anatomical Territories

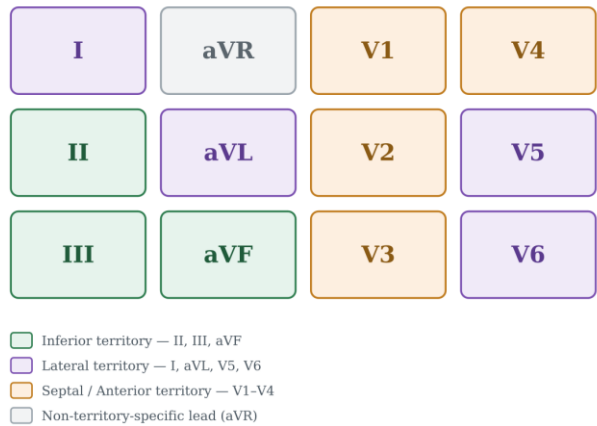


Figure 10.1 — The standard 12-lead layout, colour-coded by anatomical territory.

Territory	Lead s
Inferior	II, III, aVF
Septal / Anterior	V1–V4
Lateral	I, aVL, V5, V6

Additional leads may be required when posterior or right-ventricular involvement is suspected.

10.3 Posterior Infarction Pattern

Posterior myocardial infarction may produce reciprocal changes in the anterior leads, including ST depression in V1–V3 and relatively tall R waves anteriorly.

TIP

- Posterior leads such as V7–V9 may help evaluate suspected posterior involvement.

10.4 Right Ventricular Involvement

Right-sided leads can be useful when right ventricular infarction is suspected, particularly with an inferior infarction pattern.

TIP

- V4R is commonly used in this assessment.

10.5 STEMI and NSTEMI

STEMI

The diagnosis involves compatible clinical features plus ECG criteria for acute ST-segment elevation or equivalent patterns requiring urgent evaluation.

NSTEMI

May show ST depression, T-wave abnormalities, nonspecific changes, or occasionally no diagnostic ischemic change on the initial ECG.

Cardiac biomarkers, especially troponin, are therefore important in diagnosing myocardial infarction.

⚠ WARNING

- Critical Learning Point: a normal initial ECG does not by itself exclude acute coronary syndrome.

Chapter Summary

ECG changes in ischemia and infarction are interpreted by territory, not in isolation — inferior (II, III, aVF), septal/anterior (V1–V4), and lateral (I, aVL, V5, V6) — with posterior and right-ventricular involvement needing additional leads. STEMI and NSTEMI are distinguished by ECG criteria and biomarkers together, and a normal initial ECG never fully excludes acute coronary syndrome.

Key Takeaways

- Never diagnose infarction from ST elevation alone — location, lead count, morphology, reciprocal changes, symptoms, and serial change all matter.
- Inferior, septal/anterior, and lateral territories map to specific lead groups that should be read together.
- V7–V9 help detect posterior infarction; V4R helps detect right ventricular involvement.
- A normal initial ECG does not exclude acute coronary syndrome — troponin and clinical reassessment remain essential.

✓ QUICK RECALL

- Which leads correspond to the inferior, septal/anterior, and lateral territories?
- What additional leads help detect posterior and right ventricular infarction, respectively?
- Why can't a normal initial ECG exclude acute coronary syndrome?

CHAPTER 11

Electrolytes and Other Important Patterns

Electrolyte disturbances can alter cardiac conduction and repolarisation.

T-Wave and U-Wave Clues on the ECG

Normal



Hyperkalemia



tall, peaked, narrow T wave

Hypokalemia



flat T, prominent U wave

Figure 11.1 — Normal T wave compared with the hyperkalemia and hypokalemia patterns.

11.1 Hyperkalemia

Possible progression can include:

- Tall, narrow, peaked T waves
- PR prolongation
- P-wave flattening/disappearance
- QRS widening
- Severe conduction abnormalities

⚠ WARNING

- The ECG pattern varies with severity and individual patient factors.

11.2 Hypokalemia

Possible findings:

- T-wave flattening
- ST depression
- Prominent U waves
- Apparent QT/QU prolongation

11.3 Calcium

Abnormality	Typical QT Effect
Hypercalcemia	Tends to shorten the QT interval
Hypocalcemia	Tends to prolong the QT interval
★ KEY POINT • These patterns are clues — not substitutes for laboratory measurement.	

Chapter Summary

Potassium and calcium disturbances each leave a recognisable signature on the ECG: hyperkalemia progresses from peaked T waves toward P-wave loss and QRS widening, hypokalemia produces flat T waves and prominent U waves, and hypercalcemia/hypocalcemia shorten and prolong the QT interval respectively — but all of these are clues that should prompt laboratory confirmation, not replace it.

Key Takeaways

- Hyperkalemia's ECG progression runs from peaked T waves toward P-wave flattening and QRS widening as severity increases.

- Hypokalemia produces T-wave flattening, ST depression, and prominent U waves.
- Hypercalcemia shortens the QT interval; hypocalcemia prolongs it.
- ECG electrolyte patterns are clues that should prompt laboratory confirmation, not replace it.

✓ **QUICK RECALL**

- What is the earliest classic ECG sign of hyperkalemia?
- What two ECG findings are associated with hypokalemia besides T-wave flattening?
- How do hypercalcemia and hypocalcemia differ in their effect on the QT interval?

PART IV

Interpret Like a System

*The complete reading algorithm, common interpretation traps,
exam revision, and a rapid final checklist*

CHAPTER 12

The Complete ECG Interpretation Framework



When someone gives you an ECG, resist the temptation to jump immediately to the most dramatic-looking waveform. Use the same sequence every time.

12.1 The Twelve Steps

1. Confirm technical quality — patient/record details, paper speed, calibration, artifact, lead placement concerns.
2. Calculate the rate — choose the method according to rhythm regularity.
3. Determine rhythm — regular? irregular? irregularly irregular?
4. Inspect P waves — present? one before each QRS? uniform? relationship to QRS?
5. Measure PR — normal, short, prolonged or changing?

6. Inspect QRS — narrow or wide? normal progression? abnormal morphology? pathological Q waves?
7. Assess axis — use limb leads to estimate the frontal QRS axis.
8. Inspect ST segments — elevation? depression? which leads? are changes contiguous?
9. Inspect T waves — upright? inverted? peaked? flattened?
10. Assess QT/QTc — is repolarisation abnormally prolonged or shortened?
11. Compare related leads — look for patterns across anatomical territories rather than interpreting a single lead in isolation.
12. Form the impression.

12.2 Worked Example

Parameter	Finding
Rate	76 bpm
Rhythm	Regular sinus rhythm
P waves	Present before every QRS
PR	160 ms
QRS	90 ms
ST-T	No acute abnormality identified
Impression	Sinus rhythm

★ **KEY POINT**

- This structured format is much more reliable than jumping directly to a diagnosis.

Chapter Summary

The complete interpretation framework is the same nine-part sequence from Chapter 7, extended into twelve concrete steps that begin with a

technical quality check and end by comparing related leads before forming an impression — always in that order, never starting from the most dramatic-looking waveform.

Key Takeaways

- Always start with a technical quality check, not the waveform itself.
- The twelve steps run from rate and rhythm through to a final impression, in a fixed order.
- Lead comparison across anatomical territories comes before forming the final impression, not after.
- A structured, written-out impression (rate, rhythm, P, PR, QRS, ST-T, impression) is more reliable than an instant diagnosis.

CHAPTER 13

Common Interpretation Traps



Even with a systematic approach, certain errors recur often enough to deserve their own checklist.

13.1 The Eight Traps

Trap	Description
1. Ignoring paper speed	A tracing recorded at a different speed changes your rate and interval calculations
2. Ignoring calibration	Voltage interpretation becomes unreliable if calibration differs from standard
3. Wrong chest-lead placement	Misplaced V1/V2 or other precordial electrodes can create misleading morphology
4. Calling an ECG normal because the rate is normal	A patient can have a rate of 75 bpm and still have important conduction, ischemic or repolarisation abnormalities
5. Diagnosing from one lead	Important abnormalities generally require interpretation of the complete tracing and relevant contiguous leads
6. Treating every ST elevation as STEMI	ST elevation has several possible causes

Trap	Description
7. Assuming a normal ECG excludes disease	Some clinically significant cardiac conditions can initially have a nondiagnostic ECG
8. Ignoring QT/QTc	QT prolongation can be clinically important, particularly with certain medications, electrolyte abnormalities and congenital conditions

Chapter Summary

Most recurring ECG mistakes come from skipping a step of the systematic sequence — not checking paper speed or calibration, reading a single lead in isolation, stopping at a normal rate or a normal-looking tracing, or overlooking the QT interval. Each trap in this chapter maps directly back to a step in the twelve-step framework from Chapter 12.

Key Takeaways

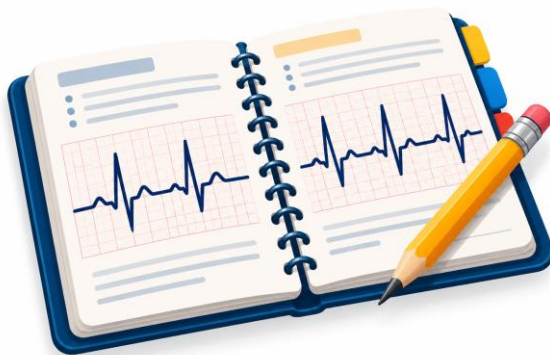
- Paper speed and calibration errors silently distort every rate and voltage measurement that follows.
- A normal rate or a single normal-looking lead does not confirm a normal ECG.
- Not every ST elevation is a STEMI, and not every normal ECG excludes disease.
- QT/QTc should never be skipped, especially when medications or electrolyte abnormalities are in play.

✓ QUICK RECALL

- Why can diagnosing from a single lead be misleading?
- Name two reasons a normal ECG does not exclude significant cardiac disease.
- Why is it a trap to treat every ST elevation as a STEMI?

CHAPTER 14

Practice, Viva & Exam Revision



Use this chapter as a rapid self-test. Cover the right-hand column and try to answer each question before checking.

Question	Answer
What does an ECG record?	Electrical activity of the heart
Standard paper speed?	25 mm/sec in the conventional setting
At 25 mm/sec, one small square represents?	0.04 sec
One large square represents?	0.20 sec
Standard voltage calibration?	10 mm = 1 mV
Normal adult resting heart-rate reference?	Approximately 60–100 bpm
What does the P wave represent?	Atrial depolarisation
What does the QRS represent?	Ventricular depolarisation
What does the T wave represent?	Ventricular repolarisation
Typical PR interval?	0.12–0.20 sec

Question	Answer
When is QRS considered wide?	≥ 0.12 sec
Which leads view the inferior territory?	II, III, aVF
Which leads view the lateral territory?	I, aVL, V5, V6
Typical ECG clue for atrial fibrillation?	No consistent sinus P waves with an irregularly irregular ventricular rhythm
First-degree AV block?	PR > 0.20 sec with 1:1 AV conduction
Mobitz I?	Progressive PR prolongation followed by a nonconducted P wave
Mobitz II?	Intermittent nonconducted P waves without progressive PR prolongation in the classic pattern
Complete heart block?	Atrial and ventricular activity are dissociated
Common hyperkalemia clue?	Tall peaked T waves can occur, with progressive conduction abnormalities in more severe cases
Common hypokalemia clue?	T-wave flattening and prominent U waves may occur

Chapter Summary

This chapter has no new content — it is a self-test drawing together the key facts from Chapters 1 through 11, formatted as question-and-answer pairs for rapid viva or exam revision.

CHAPTER 15

Rapid Reference Tables & Final Checklist



Before giving your interpretation, run through this one-page checklist.

15.1 One-Page ECG Checklist

Technical

- Correct patient/tracing
- Paper speed
- Calibration
- Artifact/lead-placement problems

Rate

- Fast / normal / slow

Rhythm

- Regular / irregular

P Waves

- Present

- Consistent morphology
- Appropriate relationship with QRS

PR

- Measure interval
- Constant or variable?

QRS

- Width
- Morphology
- R-wave progression
- Abnormal Q waves

Axis

- Normal or deviated?

ST

- Elevation
- Depression
- Distribution

T

- Morphology
- Inversion
- Peaking/flattening

QT/QTc

- Normal / prolonged / shortened

★ KEY POINT

- Final question: does the ECG fit the patient's clinical presentation?

15.2 Final Takeaway

ECG interpretation becomes easier when you stop trying to memorise hundreds of isolated patterns. Build your interpretation around three layers:

1. Understand — know what electrical event each wave represents.
2. Measure — rate, intervals, QRS width, axis and relevant ST-T changes.
3. Interpret systematically — use the same sequence on every ECG.

The goal is not to look at a tracing and instantly guess a diagnosis. The goal is to develop a repeatable method that makes you less likely to overlook an important finding.

15.3 The Core Sequence

Technical Check



Rate



Rhythm



P



PR



QRS



Axis



ST



T



QT/QTc



Lead Comparison



Final Impression